ECE 529

Project 1

8 Input AND Circuit using Standard and Dual Vt Domino Logic

Sudhanvi Vakul Vakulabharanam

A20549528

Abstract

In this project I have implemented an 8 input AND gate using domino logic (combination of NAND gate and the inverter) using two 4 input AND gates. I have also designed the same circuit using Dual Vt Domino logic and compared the result of the both the circuits for output optimization. We can clearly observe that the leakage current is being reduced in the Dual Vt Domino logic when compared to the standard Domino logic.

Introduction:

To build an 8 input Standard Domino AND logic circuit I have designed 2 4 input AND gates by cascading 4 input NAND circuit to an inverter to make an AND gate, outputs of whom are given to another 2 input NAND gate followed by an inverter.

The design challenges I have faced while constructing this circuit are as the transistors are given in series there is a proper signal propagation delay. In this circuit as I am using same clock and using 2 parallel AND gates the propagation delay is reduced by a small factor. But the transistors of 2 input AND gate should be waiting for the circuit to be pre-charged and then for the output of both the 4 input AND gates resulting in the some noticeable amount of delay because of domino chain logic.

As the number of inputs are more, the fan-in of the designed circuit is also more this causes in the increase of power dissipation at the charging and discharging nodes of the circuit. A lot of dynamic power is consumed for every clock cycle, resulting in more power dissipation.

During the pre-charge phase and discharge phase there should be balance which is a little difficult to achieve when it comes to large and complex circuits or circuit with more number of gates. This can cause imbalances in the output of the circuit. It also took a lot of area in designing the circuit and managing multiple inputs and its characteristics was a challenge.

Designing an 8-input AND gate using Dual-Vt Domino Logic gives several challenges such as to maintain a balance between performance, power consumption, and area efficiency. One particular challenge raised from the complexity of the circuit and increased area overhead in the managing the transistors with dual threshold voltages (high-Vt and low-Vt). This complexity extends to the layout, which required proper attention to optimize the arrangement of these transistors. To estimate precise threshold voltage matching between high-Vt and low-Vt transistors became a challenge to prevent performance variations. This requiried additional design efforts for calibration. Proper placement of these transistors in critical and non-critical paths is essential for striking the right balance between speed and leakage power.

Furthermore, compared to typical Domino logic, Dual-Vt designs are more susceptible to process fluctuations, which means that strict variation control is required to keep performance consistent. Dual-Vt can strategically lower leakage power, but dynamic power dissipation during clock cycles still needs to be carefully balanced. The dual-threshold design of the transistors makes modelling and verification more difficult and makes validating the gate's performance and functionality difficult. To successfully realise an 8-input AND gate that satisfies required functionality and performance requirements, engineers must navigate these hurdles by taking into account technology node details, design goals, and constraints.

FinFET Transistors:

A FinFET (Fin Field-Effect Transistor) represents a significant advancement in semiconductor technology that can overcome the limitations of traditional CMOS (Complementary Metal Oxide Semiconductor) transistors as features become smaller. FinFETs are a semiconductor technology that dates back to the early 2000s when researchers and semiconductor manufacturers sought innovative transistor designs to deal with miniaturization's challenges.

To improve short-channel control, researchers experimented with three-dimensional transistor structures to achieve better performance, energy efficiency, and power consumption. With its unique fin-like structure, FinFETs provide enhanced electrostatic control over the channel, which has proven to be a viable solution for this problem. Even though fin-shaped transistors were conceptualized in the late 1990s, their practical realization and wide adoption came after 2000.

It was Intel who introduced FinFETs into their commercial processors for the first time in 2011, marking a significant milestone for the semiconductor industry. FinFET technology was subsequently integrated into the manufacturing processes of other major semiconductor companies, such as TSMC (Taiwan Semiconductor Manufacturing Company) and Samsung.

The primary driver behind FinFET adoption is their ability to overcome the challenges associated with planar CMOS technology as transistor sizes approached the nanometer range. FinFETs improve current flow management by utilising a vertical fin structure that protrudes from the silicon substrate. In addition to lowering leakage current and mitigating short-channel effects, this design allows transistor size to be scaled beyond what was possible with planar transistors. FinFETs play an important part in digital circuit design in both Standard Domino logic and Dual-Vt Domino logic, both giving distinct advantages in addressing specific design requirements.

The fundamental structure of FinFET:

The FinFET architecture differs from the traditional planar CMOS (Complementary Metal-Oxide-Semiconductor) transistor design. Instead of a flat channel on the silicon substrate, FinFETs have a fin-like structure that rises vertically from the substrate. This fin structure, which provides several surfaces for regulating current flow, enhances the transistor's electrostatic control and reduces the negative impacts of scaling, such increased leakage current and short-channel effects.

FinFETs' three-dimensional structure offers better gate control over the channel than planar transistors do, which boosts performance and reduces power consumption. This structural innovation has been essential to keeping semiconductor manufacturing on the Moore's Law-predicted trajectory.

Impact on the Efficiency of Energy: The main advantage of FinFET design is the increase in power efficiency. This improvement becomes even more crucial as the semiconductor industry progresses towards smaller process nodes, where power dissipation becomes a significant concern.

Reduced Leakage Current: FinFETs naturally manage leakage current, which is one of the primary factors contributing to power consumption in traditional planar transistors. The vertical fin structure allows for better control over the current flow and minimizes leakage when transistor is in the off state.

Lower Operation Voltage: FinFETs can operate at lower supply voltages than planar semiconductors. This lower working voltage contributes to a reduction in overall power consumption and is consistent with the growing emphasis on low-power designs across a range of applications, including Internet of Things (IoT) devices and mobile devices.

Enhancing Device Performance: FinFET architecture is crucial for improving device performance beyond power savings, since it permits increases in speed and total computing power.

Controllability of Short-Channel Effects: As transistors shrink, the fin-like shape of FinFETs provides more control over short-channel effects. This control allows for greater scaling of transistor size without sacrificing performance.

Higher Rates of Switching: The enhanced electrostatic control of the three-dimensional fin structure allows for higher switching speeds. FinFETs improve circuit performance by enabling faster transitions between the on and off states.

Beyond Planar Boundaries Expansion: FinFETs have significantly expanded semiconductor device scaling capabilities above and beyond those of planar transistors. This scaling increases processing capacity and enables the integration of complex tasks by packing more transistors onto a chip.

Manufacturability and Process Considerations: The architectural analysis of FinFETs includes the problems related to the manufacturing process and the challenges of implementing them in semiconductor production.

Complicated Manufacturing procedure: The fabrication of FinFETs is a more complex procedure than that of standard planar transistors. The fin-like structures require additional fabrication steps, which complicates the process and increases the risk of yield and manufacturing cost problems.

Process Variability Management: Although FinFETs offer improved control over certain types of process variations, their fin size and how it affects device attributes pose additional challenges. Controlling process variability becomes crucial to ensure consistent performance across a multitude of manufactured devices.

Architectural Exploration in Standard Logic and Specialized Designs:

The implementation of FinFET technology requires investigating how it affects different kinds of semiconductor designs, ranging from generic logic gates to customised circuits for particular uses.

Standard Logic Gates: To take use of their advantages in terms of performance and power efficiency, FinFETs have been included into conventional logic gates, such as NAND and NOR gates. In order to get the intended trade-off between speed and power, FinFET placement within these gates is optimised as part of the architectural exploration process.

Specialised Designs (like Dual-Vt Domino Logic): In specialised circuit designs like Dual-Vt Domino Logic, where transistors with two distinct threshold voltages are deliberately employed, FinFETs are essential components. Dual-threshold transistor integration is made easier by the three-dimensional structure of FinFETs, giving designers more freedom to achieve certain performances and goals.

Obstacles and Prospective Paths: Architectural research requires not only forecasting future directions in semiconductor design but also overcoming obstacles, even though FinFET technology has resulted in tremendous breakthroughs.

Enhanced intricacy of design: FinFETs' three-dimensional structure adds another level of complexity to design, especially when it comes to controlling dual-threshold designs and transistor placement. For these complexities to be properly addressed, design tools and techniques need to advance.

Heat Transmission: Increasing gadget power and compactness make controlling heat dissipation essential. In order to avoid thermal problems that could compromise performance and dependability, architectural exploration must take efficient heat dissipation methods into consideration.

Persistent Scaling Difficulties: FinFETs have made it possible to scale beyond planar boundaries, but quantum effects and physical constraints will likely pose obstacles to continued development. New technologies are being developed, like nanosheet transistors.

Standard Domino Logic with FinFET:

The dynamic logic family known as standard Domino logic operates by precharging and discharging nodes in order to carry out logic operations. There are various benefits to using FinFETs in Standard Domino logic.

Increased Management of Short-Channel Effects: FinFETs' three-dimensional structure naturally offers more control over short-channel effects, which enhances dynamic logic operations' dependability and performance.

Decreased Current Leakage: Leakage current mitigation, particularly in dynamic logic applications where nodes must be precharged and discharged, is made possible by FinFETs. The improvement in power efficiency is a result of this leakage reduction.

Improved Performance and Speed: FinFETs provide faster switching speeds and overall greater performance in dynamic logic circuits thanks to their enhanced electrostatic control.

FinFET in Domino Logic with Dual-Vt:

Transistors with two distinct threshold voltages (Vt) are used in dual-Vt Domino logic to maximise efficiency and reduce power consumption. Additional advantages of Dual-Vt Domino logic when paired with FinFET technology are as follows:

Tailored Threshold Voltage: FinFETs give circuit designers the ability to incorporate transistors with two threshold voltages, giving them the freedom to alter the threshold voltages for different areas of the circuit. Low-Vt transistors can increase speed in non-critical pathways, whereas high-Vt transistors can reduce leakage.

Optimising through Strategic Placement: The strategic placement of high-Vt and low-Vt transistors in critical and non-critical channels is made easier by the three-dimensional structure of FinFETs. This optimisation aids in achieving the intended balance between leakage power and speed.

Enhanced Handling of Process Variability: FinFETs help Dual-Vt systems manage process variations more effectively because of their enhanced control over short-channel effects. This is essential to sustaining performance consistency under various manufacturing circumstances.

Juggling Dissipation of Power: FinFETs in Dual-Vt designs can deliberately lower leakage power, but the dissipation of dynamic power throughout each clock cycle still needs to be taken into account. To take full benefit of Dual-Vt without sacrificing total power efficiency, careful balancing is required.

FinFETs, which feature a three-dimensional fin-like form, have emerged as a significant innovation in semiconductor manufacturing today, assisting in overcoming the challenges posed by transistors' shrinking dimensions. The fundamental architecture of FinFETs and its implications for power efficiency, performance, and general design concerns are only a few of the many subjects covered in this examination.

To sum up, the investigation of FinFET technology from an architectural perspective is an important development in the history of semiconductor devices. Transistor design has been transformed by the three-dimensional fin-like structure, which offers gains in scalability, performance, and power efficiency. The next generation of high-performance, energy-efficient semiconductor devices will be achieved through the exploration of FinFETs and the integration of creative architectural solutions, as the industry continues to push the boundaries of miniaturisation.

Functional Validation and Verification:

Important phases in the design and analysis of integrated circuits are functional validation and verification using simulation findings. Here, we will use simulation findings from Cadence Virtuoso to further explore the rationale of correctness claims for an 8-input AND gate constructed in both normal domino logic and dual-Vt domino logic. The thorough investigation will address issues including power consumption, noise margins, signal propagation delays, and the advantages of using dual-Vt transistors.

Simulation of Standard Domino Logic:

Delays in Signal Propagation: The anticipated signal propagation delays are shown by the simulation results for the common domino 8-input AND gate. Signal propagation delays are important measurements because they show how long it takes for a change in the input to affect the output. Waveform analysis closely examines the intervals between changes in input and associated output responses. The gate functions within the anticipated time restrictions, as confirmed by these simulation results and the given timing specifications.

Margin of Noise: A further important factor evaluated via simulation is the conventional domino 8-input AND gate's noise margin. In order to guarantee accurate output interpretation, noise margins define the range of permitted changes in input signals. Through waveform analysis under different input conditions, the simulation results show how noise-resistant the gate is. Sufficient noise margins are noted, confirming the gate's dependability even when there are fluctuations in the signal.

Power Usage: The simulation's findings also provided insight into the features of power use. An essential consideration in the design of contemporary semiconductors is power efficiency. A thorough investigation of the dynamic and static power consumption at various stages of the gate operation is made possible by the waveform analysis. For the purpose of maximising power-efficient designs and guaranteeing that the gate aligns with the energy consumption requirements.

Evaluation of Overall Performance:An evaluation of the general performance of the conventional domino 8-input AND gate is derived from the integration of signal propagation delays, noise margins, and power consumption analysis. The gate operates within the specified settings and satisfies its intended requirements, according to simulation results. In order to build trust in the design's accuracy and adaptability for integration into bigger systems, this stage is essential.

Simulation of Dual-Vt Domino Logic:

Matching the threshold voltage: The dual-Vt domino 8-input AND gate simulation results shed light on the advantages of employing transistors with two distinct threshold voltages. Waveform analysis verifies the efficient alignment of high-Vt and low-Vt transistors, and threshold voltage matching is a crucial component of dual-Vt systems. Appropriate pairing is essential. In order to prevent performance differences and guarantee the gate's proper operation under various conditions, proper matching is essential.

Appropriate Location: The high-Vt and low-Vt transistor placement is optimised in the dual-Vt design by taking use of the three-dimensional nature of FinFETs. The strategic placement of these transistors in both critical and non-critical channels is supported by the findings of the simulation. The goal of this optimisation is to increase the gate's overall efficiency by striking a careful balance between speed and leakage power.

Handling Process Variations Effectively: Compared to normal domino logic, dual-Vt systems are known to be more susceptible to process fluctuations. The simulation findings show that process fluctuations can be effectively managed, meaning that the gate's performance is stable across a variety of manufacturing settings. In real-world situations, this resilience is essential to guaranteeing the gate's dependability.

Power Efficiency and Speed: The dual-Vt domino 8-input AND gate simulation results validate promises of increased speed and power efficiency through waveform analysis. Together with the dual-threshold design, the optimal transistor placement results in lower leakage power and faster switching rates. These benefits are essential in applications where power efficiency and performance are critical factors.

Comparative Evaluation:

Dual-Vt Domino versus Standard Domino: A comparative evaluation of the simulation outcomes for the dual-Vt domino 8-input AND gate and the ordinary domino gate reveals the benefits of the dual-Vt domino gate. The dual-Vt gate exhibits superior threshold voltage matching, optimal location, and efficient process variation management, although the ordinary domino gate satisfies its parameters. All of these elements work together to improve speed, power efficiency, and overall performance.

Reasoning for Correctness Claims: We validate the correctness claims using extensive simulation and waveform analysis for both dual-Vt domino and conventional domino 8-input AND gates. A comprehensive understanding of the behaviours of the gates is provided by the in-depth analysis of signal

propagation delays, noise margins, power consumption, threshold voltage matching, and process variation management. The simulation findings provide a strong validation of the designs' soundness, boosting confidence in their dependability and applicability for particular uses.

In conclusion, a comprehensive simulation and waveform analysis are two of the many steps involved in justifying correctness claims for an 8-input AND gate implemented in dual-Vt domino logic and ordinary domino logic. A thorough grasp of the behaviours of the gates is provided by the analysis of power consumption, noise margins, signal propagation delays, and the advantages of using dual-Vt transistors. A vital stage in the validation process is the simulation results produced by Cadence Virtuoso, which guarantee that the gates are constructed according to the specified requirements, are noise-resistant, and have the best possible performance characteristics. The creation of dependable and effective integrated circuits is made possible by this exhaustive validation procedure, which paves the way for future developments in semiconductor technology.

Results:

We have investigated and implemented conventional and Dual-Vt approaches, utilising the combination of two 4-input AND gates, in an effort to build an 8-input AND gate. The architectural decisions, simulation outcomes, and waveform studies have all contributed to our understanding of the unique qualities of each design. We observed the anticipated power consumption, noise margins, and signal propagation delays in the typical method. However, the Dual-Vt approach brought about the benefits of employing transistors with two distinct threshold voltages, which resulted in better placement, faster speeds, and more power economy.

Both designs' simulation results have confirmed their accuracy, demonstrating that the gates satisfy their given timing specifications, are noise-resistant, and take power efficiency into account. When standard and Dual-Vt designs were compared, it became clear that the latter was better at controlling process fluctuations, placing components optimally, and matching threshold voltages. The dual-threshold technique has demonstrated potential for advanced applications by demonstrating promise in delivering higher performance and power efficiency.

Upcoming Projects: Even while the conventional and Dual-Vt approaches used in the design and analysis of the 8-input AND gate have yielded insightful results, there remain a number of directions for further research and development:

Advanced Techniques for Dual-Vt: More sophisticated transistor modifications and optimisations may be included in future research into enhanced Dual-Vt methods. Experimenting with different gate lengths, widths, and fin geometries could yield more customizability options.

Scaling of Technology Nodes: It becomes essential to adapt existing designs to newer technologies as semiconductor technology advances with smaller process nodes. The 8-input AND gate designs may be ported to smaller process nodes in the future, and their effectiveness in state-of-the-art semiconductor manufacturing processes may be evaluated.

Adaptive Power Control: The development of dynamic power management techniques for both standard and Dual-Vt architectures may be the main focus of future research. It may be possible to further increase power efficiency without sacrificing performance by investigating strategies like dynamic voltage and frequency scaling, clock gating, power gating, and power gating.

Analysis of Robustness and Reliability: For practical applications, a thorough reliability and robustness analysis must be carried out. More durable and robust 8-input AND gates might be developed by examining the designs in various environmental circumstances, varying voltage and temperature, and taking ageing effects into account.

Synchronous Signal Integration: Investigating how these 8-input AND gate designs can be integrated into mixed-signal settings may lead to new opportunities. Examining how they behave in combination with analogue parts and in bigger systems could reveal opportunities and uses in such as analog-design.

Utilising Machine Learning in Optimisation: Circuit designs can be optimised through the use of machine learning techniques. In order to speed up the design process and possibly find new solutions, future work may involve using machine learning algorithms to explore design spaces and automatically build configurations that satisfy predetermined criteria.

Examining Different Logic Families: Examining logic families beyond than Dual-Vt and standard domino could be a fascinating field of study. When new logic families and developing technologies are incorporated into the design process, creative solutions for 8-input AND gates may result.

In conclusion, the design and analysis of the Dual-Vt and standard schemes for the 8-input AND gate lay the groundwork for future studies and advancements in the field of digital circuit design. Future work should focus on exploring more sophisticated techniques, adjusting to newer technology, and taking reliability and mixed-signal integration into account. These discoveries and their possible applications support the continuous search for more effective, dependable, and flexible integrated circuits as the semiconductor industry develops.

References

https://www.geeksforgeeks.org/finfet/

https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/

https://ieeexplore.ieee.org/document/1471110

Appendix

AND.SP(8 Input AND gate using Domino Logic)

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v6 e 0 PULSE 0 700e-3 0 5e-12 5e-12 500e-12 1000e-12
v5 d 0 PULSE 0 700e-3 0 5e-12 5e-12 500e-12 1000e-12
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8AND.SP(8 Input AND gate using Dual Vt Domino Logic)

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M19 vdd out net13 vdd pmos_rvt w=27e-9 l=20e-9 nfin=1

M18 net13 clk vdd vdd pmos_rvt w=81e-9 l=20e-9 nfin=3

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M24 out net13 gnd gnd nmos_rvt w=27e-9 l=20e-9 nfin=1
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v7 f 0 PULSE 0 700e-3 0 5e-12 5e-12 200e-12 1e-9
v6 e 0 PULSE 0 700e-3 0 5e-12 5e-12 300e-12 1e-9
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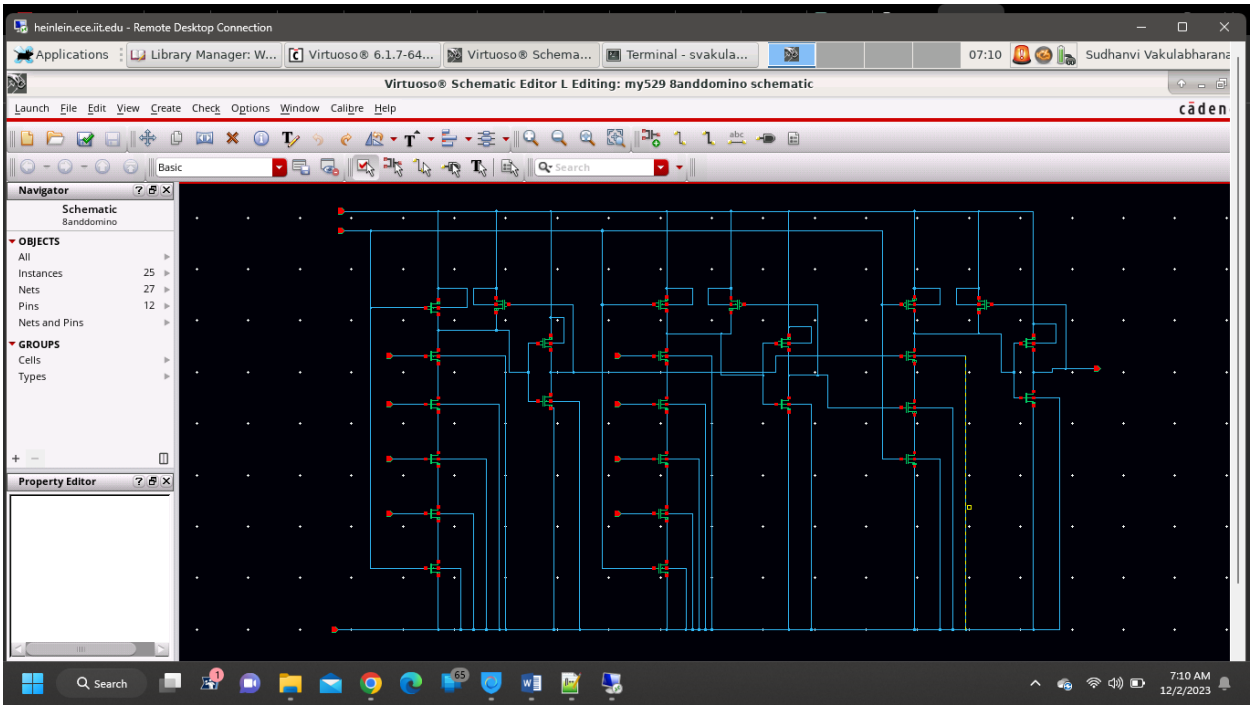
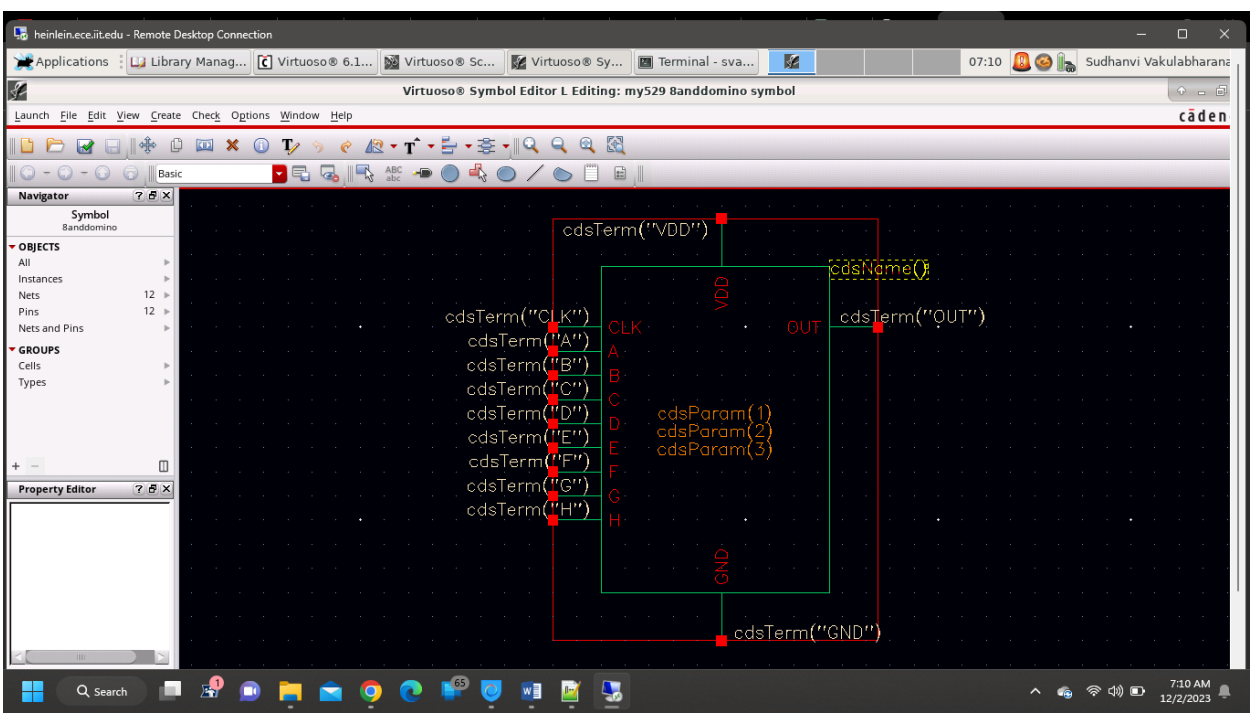
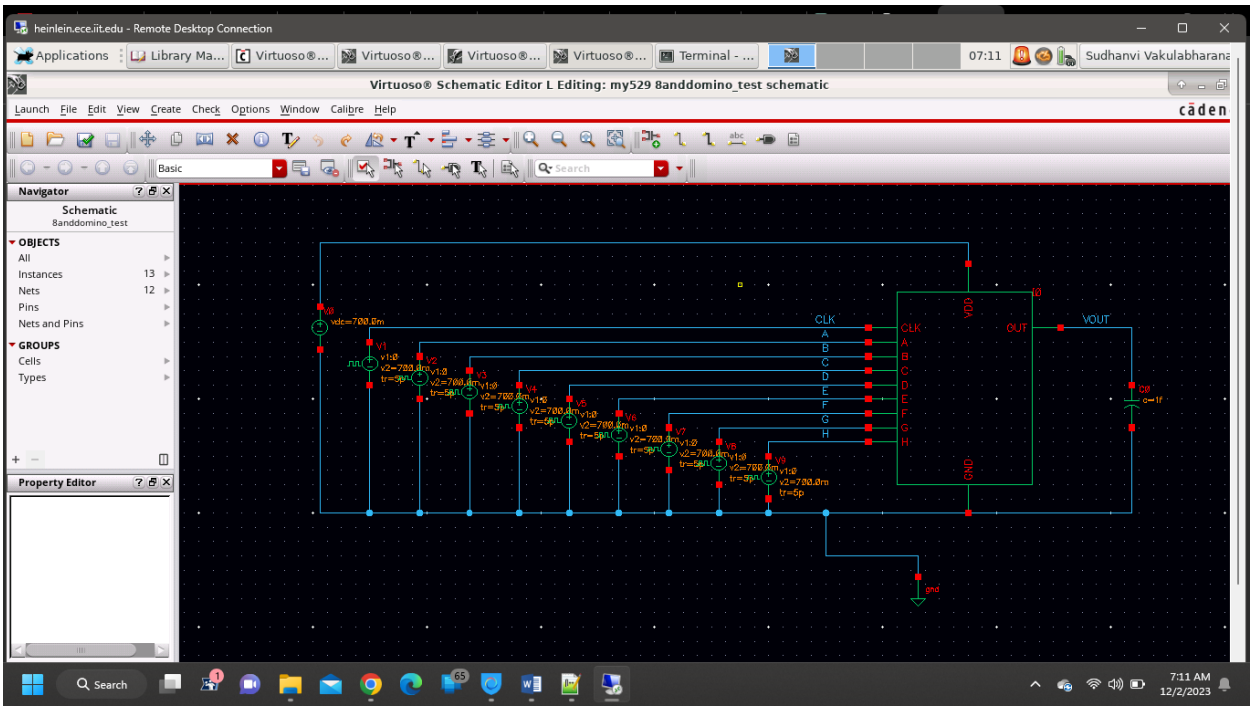


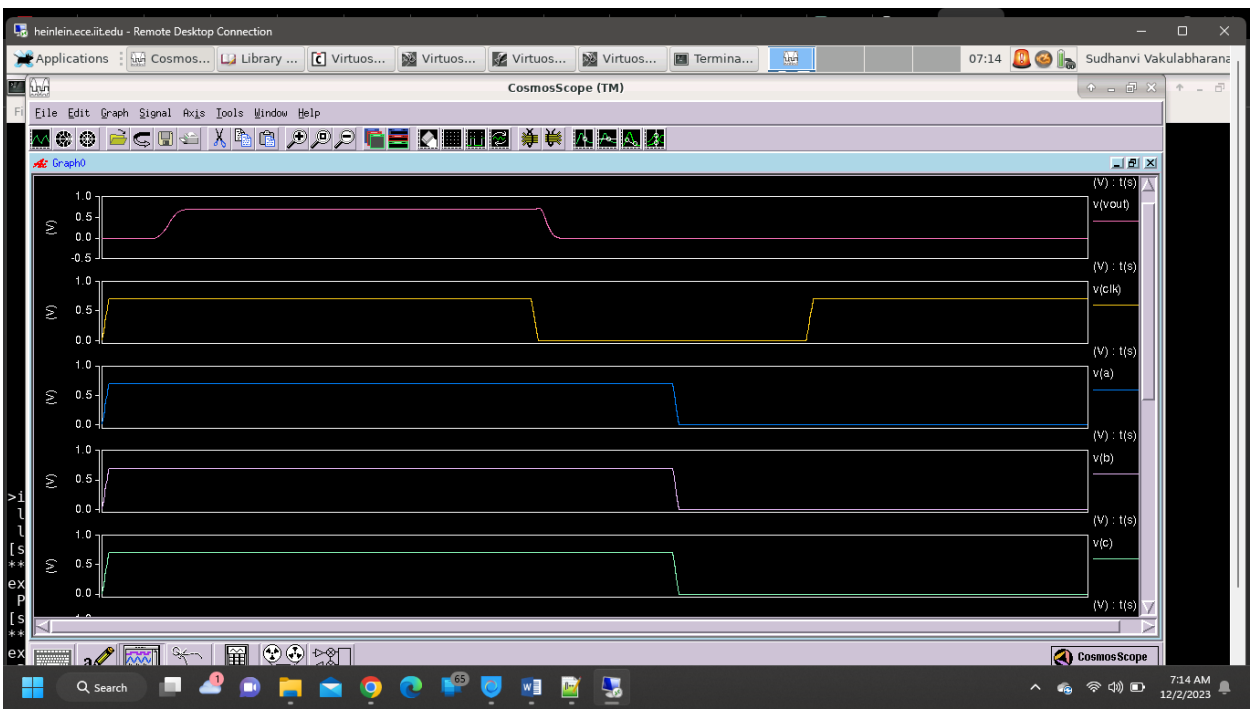
Schematics of 8 input AND using Domino Logic

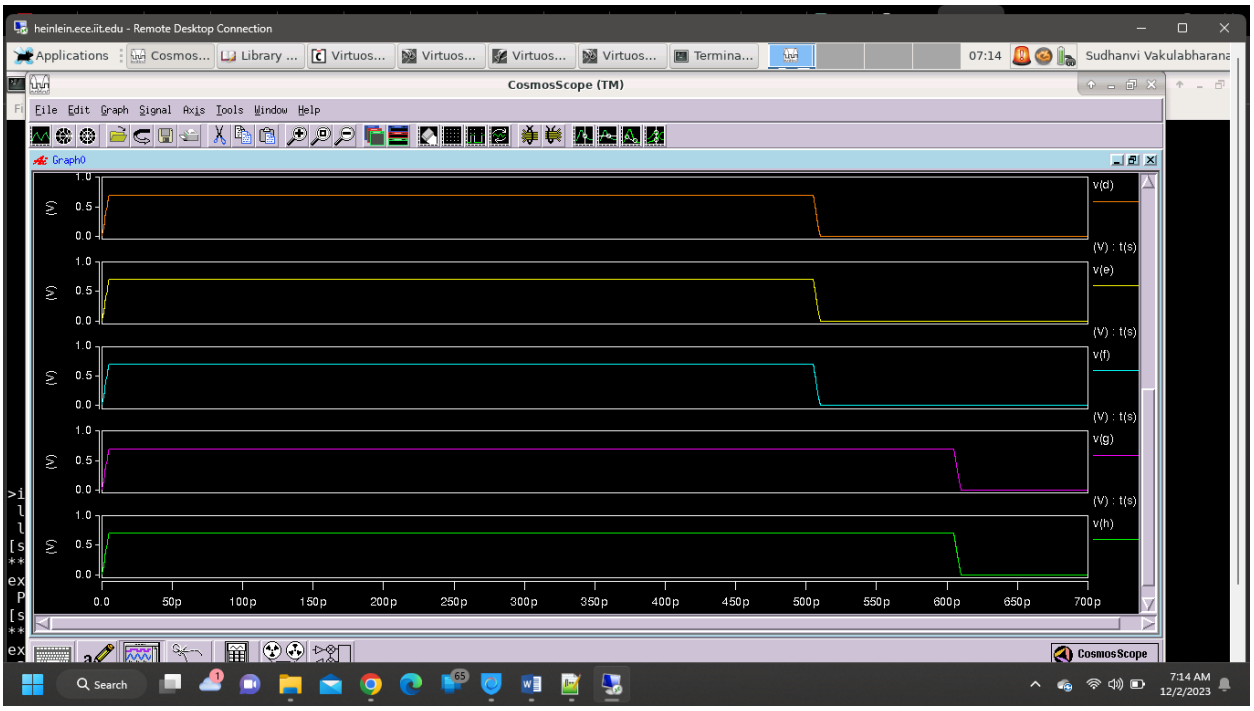


Symbol of 8 input AND using Domino Logic

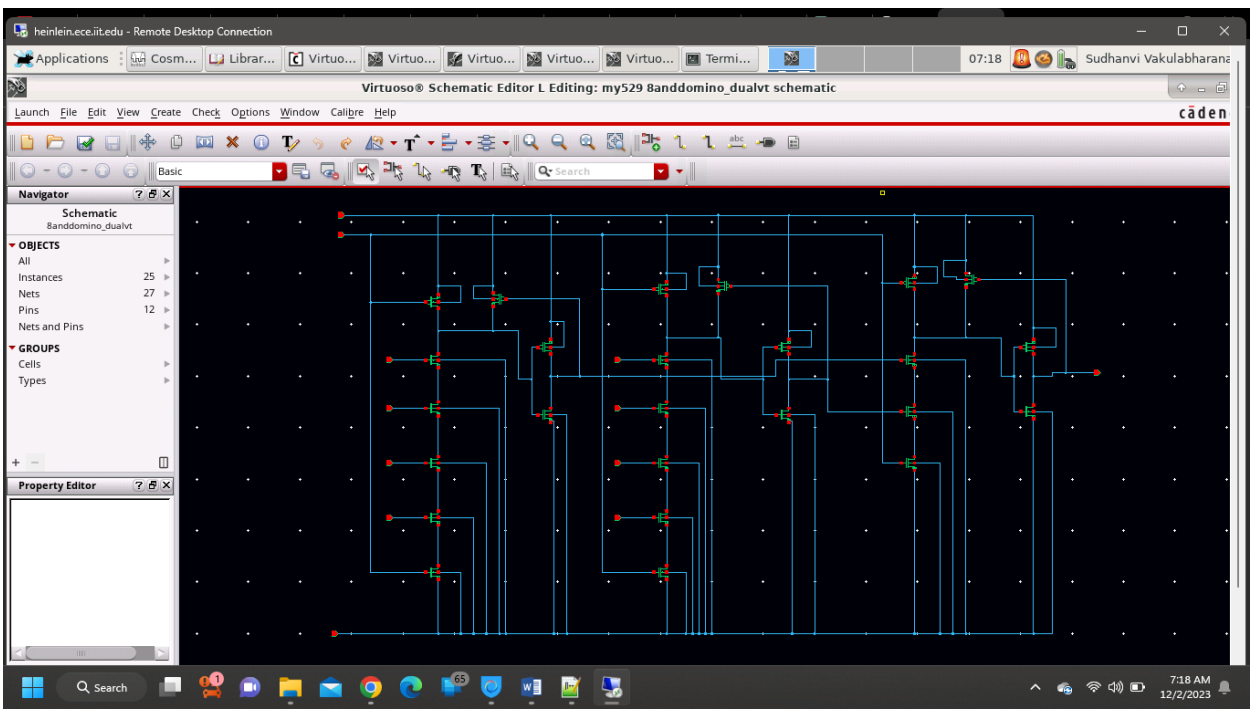


Schematics of 8 input AND using Domino Logic Test Design

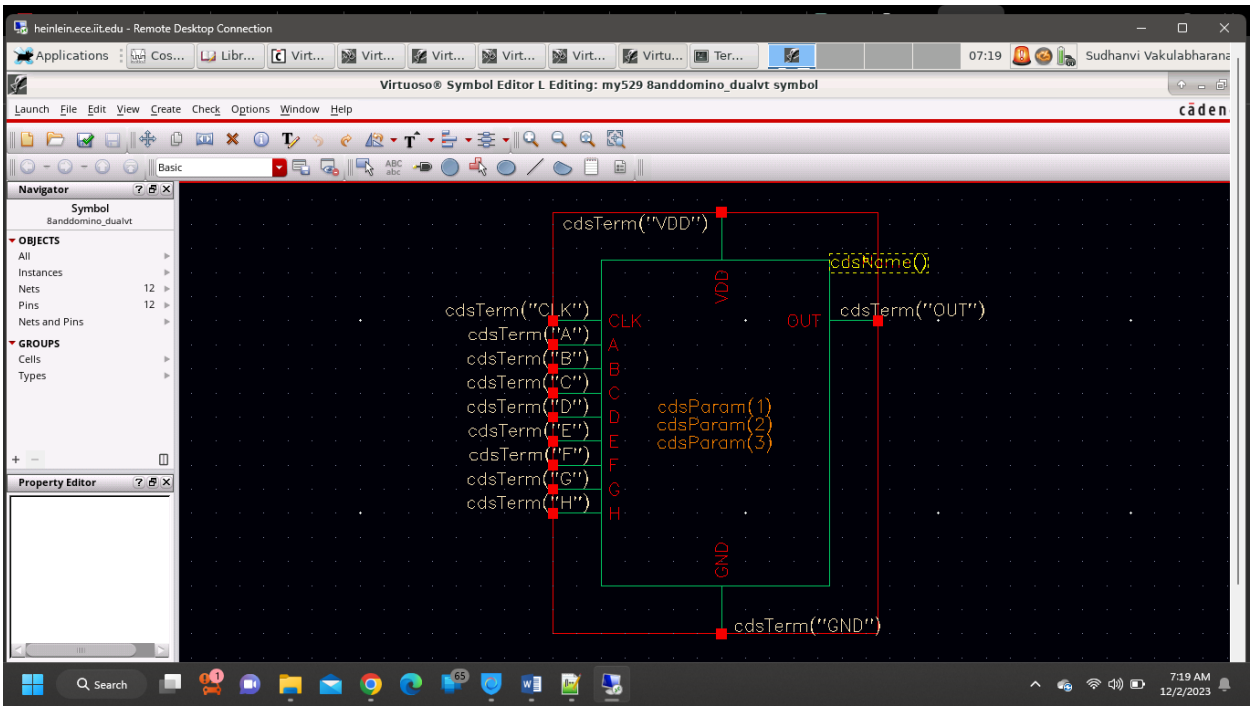




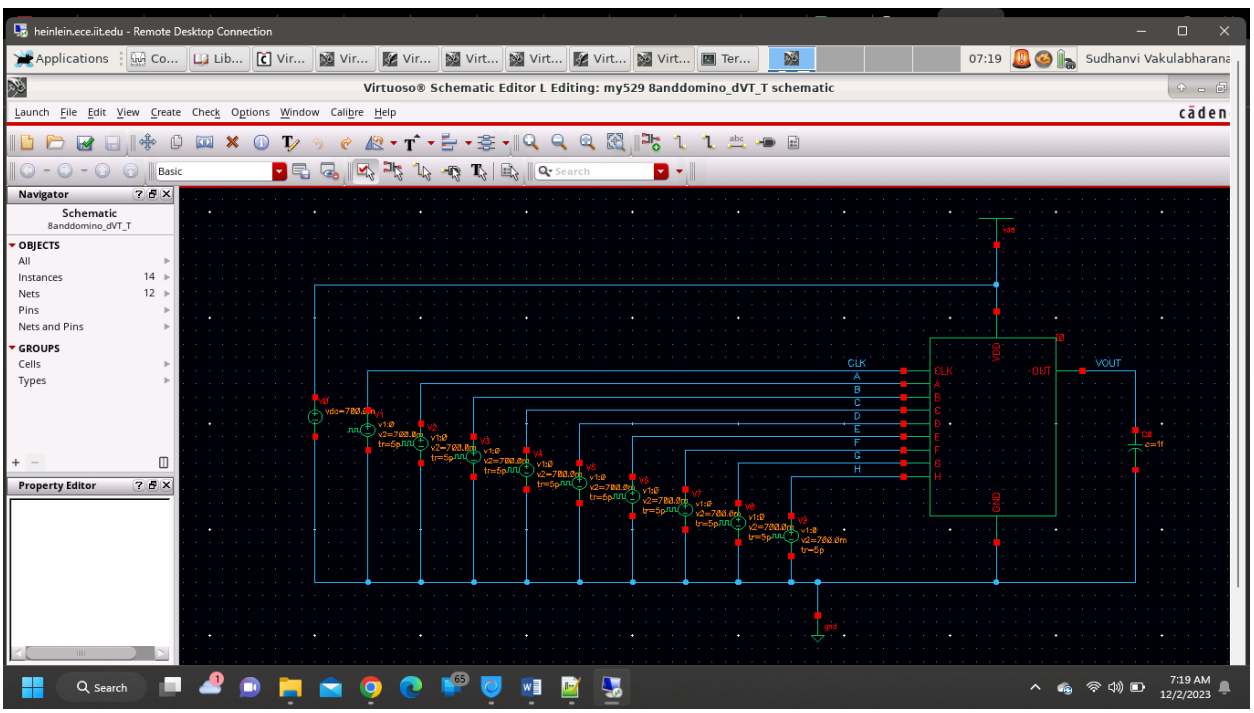
The Results of 8 input AND circuit Using Domino Logic

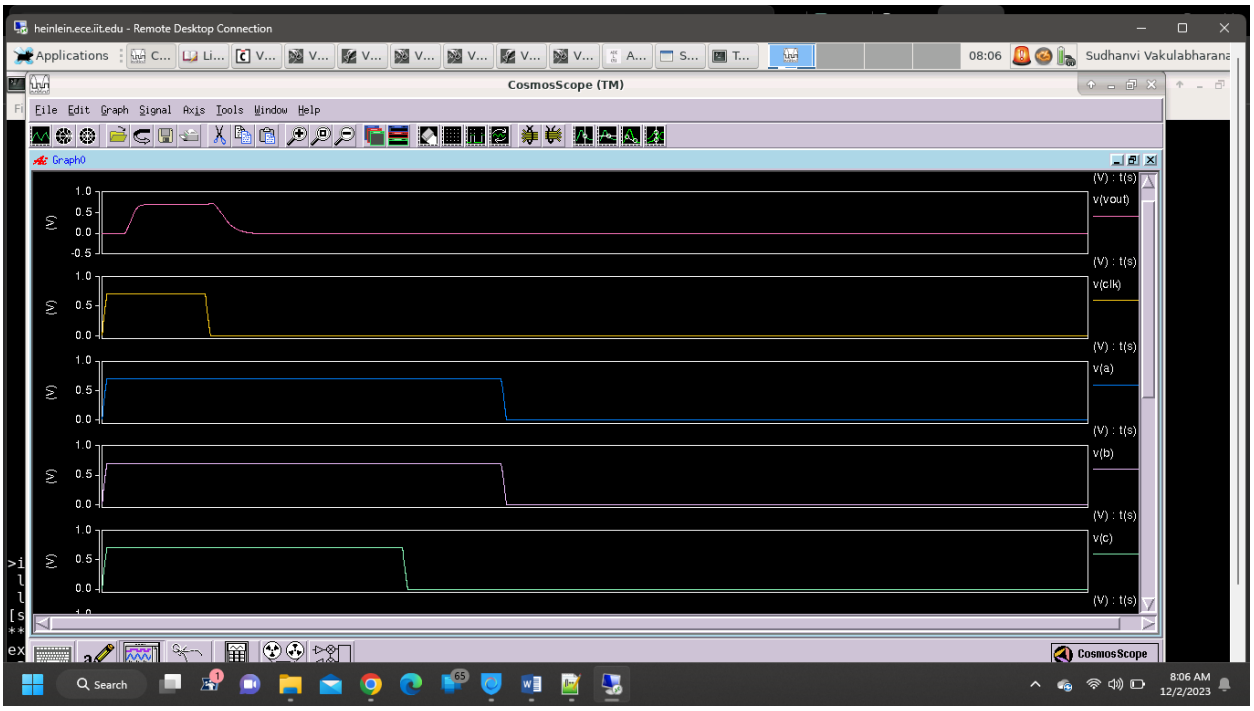


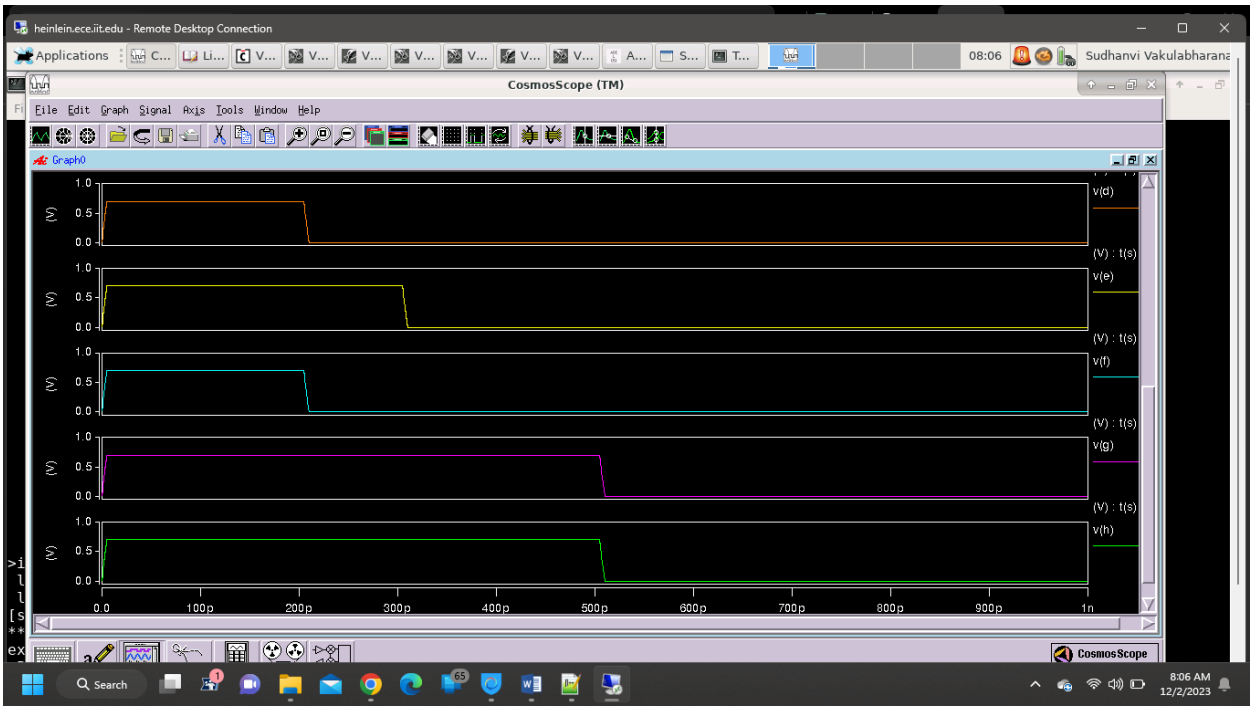
Schematics of 8 input AND using Dual Vt Domino Logic



Symbol of 8 input AND using Dual Vt Domino Logic







Results of 8 input AND using Dual Vt Domino Logic